
UNIT 10 MULTIPLE CAMERA

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10.1 INTRODUCTION

In the previous unit, we learnt about the camera models and how an image is created. We also learnt that there is loss of depth when a picture is taken by a single camera. In this unit, we shall see how we can recover the depth of a scene when there are multiple cameras. In this course, we shall study about the case when there are two cameras, that is also known as Stereo Vision. There are more limitations of a single camera system. Apart from loss of depth information, a camera is a sensor that has a fixed view volume. This implies that there is a range of scene that a single camera can see. Therefore, if the object of interest moves out of the view volume, it can no longer be observed by the camera. This may cause issues in applications like security and surveillance in wide areas. Moreover, as we saw in the study of the camera model, we can only see what lies between the camera lens and the image plane, implying that if the object of interest lies behind another object, then occlusion occurs and we are not able to view the object of interest. However, if there was another camera that could view the object of interest from another view, then we will be able to observe the object of interest despite occlusion from one view. Therefore, multiple camera systems have many advantages over single camera systems. However, this also leads to questions such as how many cameras are enough, where should the cameras be placed and how much overlap should the cameras have between their views. The answers to these questions are dependent on factors such as the cost of equipment, the type of cameras used and the application of the camera system.

10.2 OBJECTIVES

The objectives of this unit are to :

- Learn about the stereo vision system.
- Discuss the concept of point correspondences and epipolar geometry.
- Discuss the concept of motion and optical flow that allows a computer vision system to find the movement pattern in a video.

A stereo vision system consists of two cameras such that both the cameras can capture the same scene, however with some disparity between the views. One stereo vision system that we can easily relate to are the two eyes that we have.

The process of using two images of the scene captured by a stereo camera system to extract 3D information is known as stereo vision. Since the stereo pair, or the images taken by the stereo system, enable us to get the 3D information of the scene, therefore, it finds wide application in autonomous navigation of robots, autonomous cars, virtual and augmented reality, etc.

In stereo vision, the 3D information is obtained by estimating the relative depth of the points in the scene. The corresponding points in the stereo pair are matched and a disparity map is created which helps in estimating the depth of the points. We shall first understand the concept of corresponding points.

10.4 POINT CORRESPONDENCES

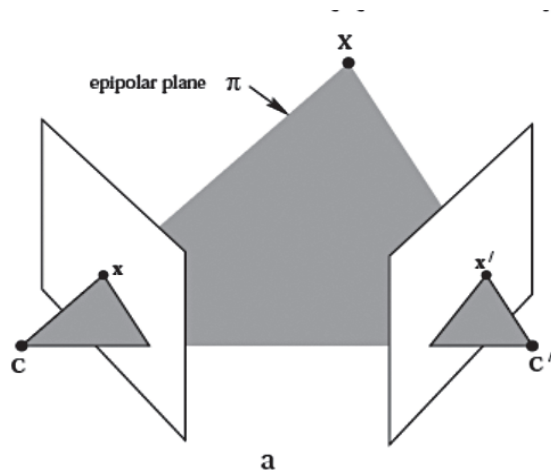


Fig10.1 The figure shows the concept of a stereo pair. C and C' are the two camera centers and x and x' are the images of the 3D point X in the corresponding image planes. x and x' are said to be corresponding points. Fig taken from [1]

As shown in Figure 10.1, the stereo pair consist of two cameras that are at a certain distance apart. The line joining the camera centers is known as the baseline. The view volume of the two cameras is such that they view a common area in the scene. A 3D point that lies in the common view volume of the two cameras will be imaged by both the cameras. Therefore, as shown in Figure 10.1, the 3D point X is visible to both the cameras and therefore, it has an image x in Camera 1 (with camera center C) and image x' in Camera 2 (with camera center C'). Therefore, x and x' are called corresponding points.

Given a point in one image, we can find its corresponding point in the other image because of epipolar geometry which we shall study next.

10.5 EPIPOLAR GEOMETRY

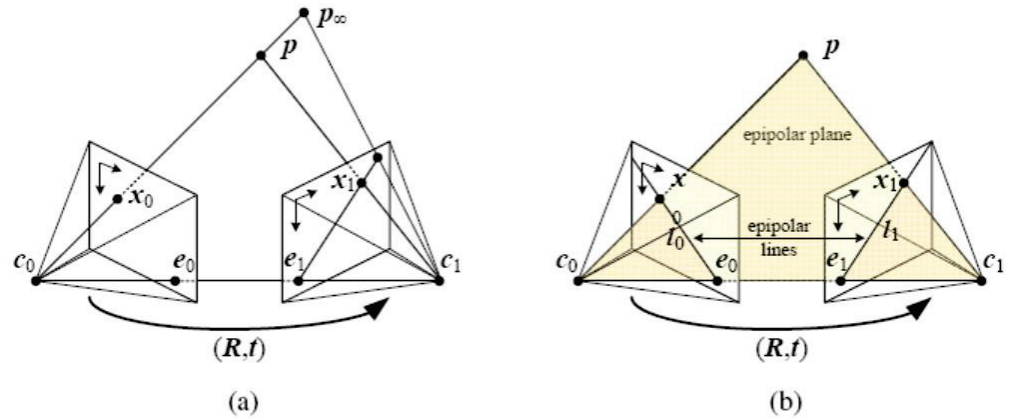


Figure 10.2. (a), (b) Epipolar geometry (Image taken from [2])

Epipolar geometry describes the geometric relation between two views of the same scene captured by a stereo vision system. In Figure 10.2 (a), it can be seen that the two cameras (C_0 and C_1) are related by a rotation (R) and translation (t), implying that the relative pose of the cameras are an inherent part of epipolar geometry. The 3D point p is imaged in both the image planes I_0 and I_1 . In I_0 , the image of p is denoted by x_0 and in I_1 , the image of p is denoted by x_1 . It can also be seen that the two camera centres, C_0 and C_1 and the 3D point p are coplanar. This plane Π is known as the Epipolar plane. Moreover, the ray joining C_0 and x_0 extends towards infinity p_∞ and is intersected by the ray joining C_1 and x_1 at the 3D point p . Therefore, we can see that in the case of stereo vision, it is possible to find the 3D point if it is imaged by both the cameras and the point correspondences are known.

The line joining the two camera centres C_0 and C_1 is known as the **baseline**. The baseline intersects the two image planes at e_0 and e_1 respectively. The point of intersection of the baseline with an image plane is known as an **epipole**. Therefore, e_0 is the left epipole and e_1 is the right epipole. The left epipole e_0 is the image of the camera center C_0 in the left image plane I_0 while, the right epipole e_1 is the image of the camera centre C_1 in the right image plane I_1 . The epipolar line is the intersection of the epipolar plane with the image plane. An important point to be noted is that every epipolar line passes through the epipole, therefore, the point of intersection of all the epipolar lines is the epipole. Another point to be noted is that the epipolar

line $\mathbf{l}_1$ in I_1 is the image of the back projected ray joining the camera centre $\mathbf{C}_0$ with the 3D point $\mathbf{p}$, while the epipolar line $\mathbf{l}_0$ in I_0 is the image of the back projected ray, that is the ray from the camera centre $\mathbf{C}_1$ with the 3D point $\mathbf{p}$.

Therefore, given $\mathbf{x}_0$, the corresponding point $\mathbf{x}_1$ in the second image is constrained to lie on the corresponding epipolar line. The Fundamental matrix, $\mathbf{F}$, represents the epipolar geometry algebraically. An important point to be noted is that the fundamental matrix defines the relation between corresponding points as given by Equation 10.1.

$$\mathbf{x}_1^T \mathbf{F} \mathbf{x}_0 = 0 \quad (10.1)$$

More precisely, the fundamental matrix $\mathbf{F}$ maps a given point $\mathbf{x}_0$ from the first image to the epipolar line $\mathbf{l}_1$ in the second image, $\mathbf{l}_1 = \mathbf{F} \mathbf{x}_0$ and since the corresponding point $\mathbf{x}_1$ of $\mathbf{x}_0$ in the second image lies on $\mathbf{l}_1$ implies Equation 10.1 is satisfied.

Since $\mathbf{x}_1^T \mathbf{l}_1 = 0$, and $\mathbf{l}_1 = \mathbf{F} \mathbf{x}_0$ therefore, $\mathbf{x}_1^T \mathbf{F} \mathbf{x}_0 = 0$.

$\mathbf{F}$ is a matrix of size 3×3 with rank 2 and therefore, if we have at least 8 correspondences, then $\mathbf{F}$ can be computed from the point correspondences.

An important property of the fundamental matrix is that $\mathbf{l}_1 = \mathbf{F} \mathbf{x}_0$ and $\mathbf{l}_0 = \mathbf{F}^T \mathbf{x}_1$. Moreover, since the epipole lies on the epipolar lines, therefore, $\mathbf{F} \mathbf{e}_0 = 0$ and $\mathbf{e}_1^T \mathbf{F} = 0$.

Solving for the Fundamental Matrix

Given point correspondences, $\mathbf{x}_i$ and $\mathbf{x}_i'$ we can setup a system of equations using Equation (1) to solve for $\mathbf{F}$.

$\mathbf{x}_i'^T \mathbf{F} \mathbf{x}_i = 0$ implies

$$[x_i' \ y_i' \ 1] \begin{pmatrix} f_{10} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{pmatrix} \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix} = 0 \quad (10.2)$$

which gives Equation 10.3 on solving:

$$x_i' f_{10} x_i + x_i' f_{12} y_i + x_i' f_{13} + y_i' f_{21} x_i + y_i' f_{22} y_i + y_i' f_{23} + f_{31} x_i + f_{32} y_i + f_{33} = 0$$

$$(10.3)$$

If we consider m such correspondences, then we can solve the system of equation for the 9 unknowns in F .

$$\begin{bmatrix} x_1 x'_1 & x_1 y'_1 & x_1 & y_1 x'_1 & y_1 y'_1 & y_1 & x'_1 & y'_1 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_m x'_m & x_m y'_m & x_m & y_m x'_m & y_m y'_m & y_m & x'_m & y'_m & 1 \end{bmatrix} \begin{bmatrix} f_{11} \\ f_{21} \\ f_{31} \\ f_{12} \\ f_{22} \\ f_{32} \\ f_{13} \\ f_{23} \\ f_{33} \end{bmatrix} = 0 \quad (10.4)$$

This system requires atleast 8 points to solve since F has 8 degrees of freedom. Therefore, this is also known as the 8-point algorithm.

We use Singular Value Decomposition (SVD) to solve the system of equations in Equation 10.4. If the cameras are calibrated and the matrices K and K' of internal parameters are known then, we can use the fundamental matrix, F , to compute the essential matrix, E , given by Equation 10.5.

$$E = K' F K \quad (10.5)$$

The Essential matrix, E , is used to determine the camera pose i.e. the positioning and alignment of camera.

10.6 MOTION

In our daily lives, we perceive, understand and predict motion rather easily. Motion perception is very strong in the human vision system. Motion or perception of motion during the imaging process can be caused by some of the following reasons:

1. The camera is static but the object is moving.
2. The camera is moving but the object is static.
3. Both the camera and objects are moving
4. The camera is static but light source and objects are moving.

Motion perception plays an important role in applications of computer vision such as activity recognition, surveillance, 3D reconstruction, and many others. Therefore, it is important to estimate the motion from images.

10.6.1 Optical Flow

Estimating motion of the pixels in a sequence of images or videos has a very large number of applications. Optical flow is used to compute where the motion information between sequence of images. It helps to determine a dense point to point correspondence between pixels of an image at time t

with the pixels in the image at time $t+1$. Therefore, optical flow is said to compute the motion in a video or sequence of images.

Optical flow is based on the assumption that across consecutive frames, the pixel intensities do not rapidly change. It also assumes that neighbouring pixels have similar pattern of motion. Most often, it also assumes that the luminance remains constant. We assume that $f(x,y,t)$ is a pixel in the image taken at time t , moves to a point $(x+\delta x, y+\delta y)$ at the time $t+\delta t$, that is, to $f(x+\delta x, y+\delta y, t+\delta t)$. Since they are the same point, and therefore, the above assumption can be written mathematically as

$$f(x + \delta x, y + \delta y, t + \delta t) = f(x, y, t) \quad (10.6)$$

Equation (10.6) forms the basis of the 2D motion constraint equation and holds true given that $\delta x, \delta y, \delta t$ are small. Taking into consideration the first order Taylor series expansion about $f(x, y, t)$ in Equation (10.6), we get

$$f(x + \delta x, y + \delta y, t + \delta t) = f(x, y, t) + \frac{\partial f}{\partial x} \delta x + \frac{\partial f}{\partial y} \delta y + \frac{\partial f}{\partial t} \delta t + (\text{Higher order terms}) \quad (10.7)$$

Assuming that the higher order terms are very small, we ignore them. Then, from Equations (3) and (4), we get:

$$\begin{aligned} \frac{\partial f}{\partial x} \delta x + \frac{\partial f}{\partial y} \delta y + \frac{\partial f}{\partial t} \delta t &= 0 \\ \text{or, } \frac{\partial f}{\partial x} \frac{\delta x}{\delta t} + \frac{\partial f}{\partial y} \frac{\delta y}{\delta t} + \frac{\partial f}{\partial t} &= 0 \\ \text{or, } \frac{\partial f}{\partial x} v_x + \frac{\partial f}{\partial y} v_y + \frac{\partial f}{\partial t} &= 0 \end{aligned} \quad (10.8)$$

where, $v_x = \frac{\delta x}{\delta t}$, $v_y = \frac{\delta y}{\delta t}$ are the image velocities or optical flow at (x, y) at time t .

$\frac{\partial f}{\partial x} = f_x$; $\frac{\partial f}{\partial y} = f_y$; $\frac{\partial f}{\partial t} = f_t$ are the image intensity derivatives at (x, y) .

Then, from Equation (10.8), we get

$$(f_x, f_y) \cdot (v_x, v_y) = -f_t \quad (10.9)$$

Equation (10.9) is called the 2D Motion Constraint Equation.

There are various methods for solving for optical flow. The Lucas and Kanade optical flow algorithm [3] and Horn and Shunck [4] optical flow methods are two of the most popular methods for estimating the optical flow.

10.7 SUMMARY

In this unit we learned about various concepts related to Multiple camera models required for computer vision. The concept of stereo vision was discussed and the concept was extended to point correspondence and Epipolar geometry; finally the unit was completed with the motion oriented concepts which includes optical flow.

10.8 QUESTIONS AND SOLUTIONS

Q1. What is Stereo Vision ?

Sol. Ref. 10.3

Q2. Describe the concept of Point Correspondance.

Sol. Ref. 10.4

Q3. Discuss the term Epipolar Geometry.

Sol. Ref. 10.5

Q4. Explain Optical flow, in context of motion perception in computer vision.

Sol. Ref. 10.6

10.9 REFERENCES

- [1] Hartley and Zisserman “Multiple View Geometry, Oxford Press
- [2] Computer Vision: Algorithms and Applications by Richard Szeliski
- [3] B. D. Lucas and T. Kanade, “An Iterative Image Registration Technique with an Application to Stereo Vision”, DARPA Image Understanding Workshop, 1981, pp 121-130. (also IJCAI’81, pp674-679)
- [4] B. K. P. Horn and B. G. Shunck “Determining Optical Flow”, Artificial Intelligence 17, 1981, pp. 185- 204.